

# Zhuohua Liu

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Date of Birth: March 2003



## Education

### Bachelor of Engineering

- School of Mechanical, Electrical and Control Engineering, **Shenzhen University** Sept 2021 – Jul 2025
- Class Rank: Top 10%    Weighted Average Score: 88.2 / 100    GPA: 3.97 / 4.50

### Ph.D. Candidate

- School of Integrated Circuits Science and Engineering, **Beihang University** Sept 2025 – Present
- Expected graduation: 2030



## Research Experience

### Publication Record

1× DAC, 2× ICCAD, 2× ASP-DAC, 1× ISEDA, 1× ICCD, 1× CCF DAC

### KATO: Knowledge Alignment and Transfer for Transistor Sizing Of Different Design and Technology

- Project:** Proposed an efficient and generalizable knowledge alignment and transfer framework for analog transistor sizing, reducing simulation cost by up to 50% and improving design performance by up to 20%.
- Contribution:** Designed the cross-circuit knowledge transfer framework, reproduced SOTA baselines (MACE, SMAC, USEMOC, MESMOC, etc.), and contributed to writing the proposed method and experimental design sections.
- Outcome:** Paper accepted to **DAC** (CCF A, student 2nd author)

### DIVE: Dynamic Information-Guided Variable Expansion for Deeper Analog Circuit Optimization

- Project:** Proposed DIVE, a dynamic information-guided variable expansion framework that tackles the dimensionality challenge in analog circuit optimization, achieving up to 21.11× simulation reduction and 2.69× performance improvement.
- Contribution:** Conceived the core idea, designed and implemented the algorithmic framework, and led the paper writing.
- Outcome:** Paper accepted to **ICCAD** (CCF B, 1st author)

### OpenYield: An Open-Source SRAM Yield Analysis and Optimization Benchmark Suite

- Project:** Built a realistic SRAM yield analysis and optimization benchmark integrating parasitic-aware circuit generation and baseline evaluation frameworks.
- Contribution:** Led the transistor-sizing optimization and FoM evaluation pipeline.
- Outcome:** Paper accepted to **ICCD** (CCF B, **Best Paper Candidate**, co-1st author)

### DSFold: A “Free-lunch” Add-on for Efficient Design Convergence in Transistor Sizing

- Project:** Developed a framework that enhances existing Bayesian optimization methods for transistor sizing, reducing simulation cost by up to 50% and improving design performance by up to 50%.
- Contribution:** Designed and implemented the BO-enhancement framework, reproduced SOTA baselines (REMBO, cBO, BatchBO, etc.), and contributed to paper writing.
- Outcome:** Paper accepted to **ISEDA** (EI, 1st author)



## Competitions & Awards

National Scholarship

Nov 2024

2025 CHINA POSTGRADUATE IC INNOVATION COMPETITION

EDA ELITE CHALLENGE CONTEST

Nov 2025

- 1st Prize, National Final.